Amal S. Sebastian

Curriculum Vitae

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Education

2023-Present **Ph.D.**, *University of Michigan, Ann Arbor*, Aerospace Engineering and Scientific Computing.

2022–2023 **Ph.D. (transferred to the University of Michigan)**, *Carnegie Mellon University*, Mechanical Engineering.

2017–2022 **Master of Technology and Bachelor of Technology**, *Indian Institute of Technology Bombay*, Aerospace Engineering.
Minor in Computer Science and Engineering

Awards, Scholarships and Achievements

2024 30,000 A100 GPU hours by the Nvidia academic compute grant.

2019-2024 Inducted as a student member of the Indian National Academy of Engineering.

2015 Recipient of the prestigious National Talent Search Examination (NTSE) scholarship given by NCERT.

Publications

2026 **Sebastian, A. S.**, Kaneko, S., Hajdik, H., Uranga, A., Martins, J. R., & Viswanathan, V. (2026). Battery Modeling To Enable Gradient-Based Optimization of Electrified Aircraft Mission and Design. In AIAA AVIATION 2026 Forum. American Institute of Aeronautics and Astronautics. AIAA AVIATION 2026 Forum. `https://doi.org/10.2514/6.2026-4539`

2026 Huang, C., **Sebastian, A. S.**, & Viswanathan, V. (2026). Learning second-order total variation diminishing flux limiters using differentiable solvers. Physics of Fluids, 38(3). `https://doi.org/10.1063/5.0311710`

2025 Yao, A. M., **Sebastian, A.**, & Viswanathan, V. (2025). Unifying Chemical and Electrochemical Thermodynamics of Electrodes. arXiv. `https://doi.org/10.48550/ARXIV.2507.10677`

2024 Madhavan, V.*, **Sebastian, A. S.***, Ramsundar, B., & Viswanathan, V. (2024). Self-supervised pretraining for partial differential equations. arXiv. `https://doi.org/10.48550/arxiv.2407.06209`

2021 Ramachandran, P., Bhosale, A., Puri, K., Negi, P., Muta, A., Dinesh, A., Menon, D., Govind, R., Sanka, S., **Sebastian, A. S.**, Sen, A., Kaushik, R., Kumar, A., Kurapati, V., Patil, M., Tavker, D., Pandey, P., Kaushik, C., Dutt, A., & Agarwal, A. (2021). PySPH: A Python-based Framework for Smoothed Particle Hydrodynamics. ACM Transactions on Mathematical Software, 47(4), 1–38. `https://doi.org/10.1145/3460773`

* - equal contribution

Projects

2025 - Present **Battery modeling for conceptual aircraft design**.

- Integrated a hierarchy of lithium-ion battery models, from constant-efficiency sizing models to equivalent circuit models and physics-based electrochemical SPMe, DFN models using PyBaMM, into NASA's Aviary code for gradient-based electric-aircraft mission optimization.
- Implemented a common power-input battery interface returning state of charge, voltage, current, and heat generation, enabling coupled propulsion, electrical-distribution, and thermal-management sizing studies.
- Quantified aircraft-level tradeoffs between energy- and power-oriented cells, showing that climb-power requirements rather than stored energy determine battery-pack sizing.
- Demonstrated that voltage- and current-aware battery models are required to capture cable sizing, electrical-distribution mass, and their feedback on aircraft weight and mission energy.

2024 **Neural ODE battery surrogate**.

- Developed a Neural ODE–based battery degradation surrogate to model state-of-health evolution in satellite batteries, combining data-driven learning with physically structured dynamics.
- Applying the model to identify accelerated aging protocols that preserve degradation trajectories while reducing experimental time and cost.

2023 - 2024 **Self supervised pretraining for partial differential equations**.

- Trained Vision Transformer (ViT) and SwinUNet-style architectures on the PDEBench dataset to develop autoregressive neural emulators for nonlinear PDE systems.
- Demonstrated stable autoregressive rollouts and generalization within the interpolation regime, while identifying limitations in out-of-distribution and cross-regime settings.
- Scaled training using multi-GPU distributed paradigms and performance profiling to mitigate I/O and data pipeline constraints.
- Supported by an NVIDIA Academic Compute Grant (30,000 A100 GPU hours).

2022 - 2023 **Co-Administrator of the Arjuna Supercomputing Cluster**.

23 GPU nodes, 52 CPU nodes

- Managed operations and maintenance of a heterogeneous HPC cluster supporting multi-user scientific workloads
- Supported software deployment and environment management using Spack, including resolving dependency and build issues.

2019 **PySPH and Mayavi**.

- Contributed to the open-source Smoothed Particle Hydrodynamics framework **PySPH**, focusing on free-surface and multiphase flow simulations.
- Designed and implemented simulations of sloshing tank and dam-break problems, demonstrating agreement with experimental data and establishing reusable benchmark configurations.
- Re-implemented several surface tension formulations from published SPH literature, analyzing their numerical behavior, stability, and interface fidelity.
- Benchmarked models on challenging interfacial flows (droplet deformation, bubble dynamics, Kelvin–Helmholtz instability), identifying trade-offs between accuracy and robustness.
- Extended the visualization and preprocessing workflow via **Mayavi** tooling for particle generation and geometric inspection.